MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **PERIODIC ASSESSMENT TEST – 11 (QUADRATIC EQUATIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 13(iii).
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions (VSAQ) of 2 marks each.
4. **Section C** comprises 2 Short Answer questions (SA) of 3 marks each.
5. **Section D** comprises 2 Long Answer questions (LA) of 5 marks each.
6. **Section E** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A (Multiple Choice & Assertion-Reason Questions 6 × 1 = 6 Marks)

Q1. If the quadratic equation $x^2 + k(4x + k - 1) + 2 = 0$ has equal roots, then $k =$: [1]

(A) $-\frac{2}{3}, 1$ (B) $\frac{2}{3}, -1$ (C) $\frac{3}{2}, \frac{1}{3}$ (D) $-\frac{1}{3}, \frac{2}{3}$

Q2. If the roots of the equation $(a^2 + b^2)x^2 - 2b(a + c)x + (b^2 + c^2) = 0$ are equal, then: [1]

(A) $2b = a + c$ (B) $b = \frac{2ac}{a+c}$ (C) $b^2 = ac$ (D) $b = ac$

Q3. If the equation $ax^2 + 2x + a = 0$ has two equal roots, then $a =$: [1]

(A) ± 1 (B) ± 2 (C) 0 (D) $\pm\frac{1}{2}$

Q4. If the quadratic equation $x^2 - bx + 1 = 0$ does not possess real roots, then: [1]

(A) $-3 < b < 3$ (B) $-2 < b < 2$ (C) $b > 2$ (D) $b < -2$

Q5. If a and b are the roots of the quadratic equation $x^2 + ax + b = 0$, then $a + b =$: [1]

(A) -1 (B) 1 (C) 2 (D) -2

Q6. Assertion-Reason: [1]

Directions: (A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.

Assertion (A): A quadratic equation cannot possess more than two real roots.

Reason (R): Fundamental Theorem of Algebra guarantees that a polynomial of degree n has at most n roots.

SECTION B (Very Short Answer Questions 2 × 2 = 4 Marks)

Q7. Solve the following quadratic equation for x by factorization: [2]

$$2x^2 + ax - a^2 = 0$$

Q8. Find the roots of the following quadratic equation: [2]

$$x^2 - 2ax + (a^2 - b^2) = 0$$

SECTION C (Short Answer Questions $2 \times 3 = 6$ Marks)

Q9. The sum of two numbers is 15. If the sum of their reciprocals is $\frac{3}{10}$, find the two numbers. [3]

Q10. A two-digit number is four times the sum of its digits and twice the product of its digits. Find the number. [3]

SECTION D (Long Answer / HOTS Questions $2 \times 5 = 10$ Marks)

Q11. Two pipes running together can fill a cistern in $3\frac{1}{13}$ minutes. If one pipe takes 3 minutes more than the other to fill it, find the time in which each pipe would fill the cistern separately. [5]

Q12. At t minutes past 2 pm, the time needed by the minute hand of a clock to show 3 pm was found to be 3 minutes less than $\frac{t^2}{4}$ minutes. Find the value of t. [5]

SECTION E (Applied Case Study Problem $1 \times 4 = 4$ Marks)

Q13. Case Study: Railway Logistics & Freight Corridor Speed Optimization

A dedicated freight corridor connects an inland manufacturing cluster to an international maritime port located 360 km away. A heavy container cargo train travels at a scheduled average velocity of v km/h. Logistics engineers compute that increasing the average velocity by 5 km/h would save exactly 1 hour on the 360 km transit.

(i) Write algebraic expressions for transit time at velocity v and velocity $(v + 5)$ over the 360 km track. [1]

(ii) Formulate the quadratic equation in v in standard form representing the transit time reduction. [1]

(iii) Determine the scheduled average velocity of the container train (v). [2]

OR

Calculate the total operating time saved per month if the freight train operates 60 scheduled runs at the optimized higher velocity.

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